

EpiKG: End-to-End Epistemic Preservation in Clinical Knowledge Graphs for Assertion-Aware Retrieval-Augmented Generation

Anonymous Author(s)

Abstract

Clinical NLP pipelines detect assertion status—negation, uncertainty, family history—with $F_1 > 0.90$ [1], yet no existing system propagates these labels through knowledge graph construction into retrieval-augmented generation. We call this the *epistemic propagation gap*: when assertion labels are discarded during graph construction, downstream generation cannot faithfully represent the epistemic state of clinical findings. We present EpiKG, a clinical knowledge graph that closes this gap with a 7-value assertion schema on every edge, a tri-temporal model (valid time, transaction time, and NLP-asserted temporality), and an intent-aware retrieval layer that dispatches change, current-state, and historical questions to specialized graph-traversal strategies. On ClinicalBench ($n = 400$, MIMIC-IV, MedGemma 27B), EpiKG unlocks reasoning categories that are otherwise inaccessible: change-detection accuracy rises from 0% to 60% (+60 pp), family-history from 0% to 57% (+57 pp), and current-state from 64% to 76% (+12 pp). These gains are concentrated: in aggregate, intent-aware KG-RAG does not lift overall accuracy (−5.5 pp), because retrieval introduces noise in categories where the LLM is already strong. Notably, vanilla RAG *without* epistemic structure drops accuracy by 9.0 pp—assertion-blind retrieval is strictly worse than no retrieval. These results establish that the right evaluation lens for clinical KG-RAG is the category×condition interaction, not the aggregate, and that epistemic structure is the difference between retrieval that helps and retrieval that harms.

1 Introduction

A physician documents “patient denies chest pain, sister had MI at 45, will consider statin if lipids remain elevated”—a single sentence encoding four epistemic states: a negated symptom, a family-history event, a hypothetical action, and an implicit present condition. Clinical NLP can detect these assertions accurately [2, 1], yet when a RAG system ingests the same note, assertion context is flattened: negation is lost, family attribution collapses onto the patient, and downstream reasoning may conclude the patient *has* chest pain and *had* an MI. We call this the *epistemic propagation gap*—the systematic loss of assertion status as clinical information moves from extraction through storage to retrieval.

The gap persists because no system, to our knowledge, propagates assertion labels beyond the extraction stage into knowledge representation or retrieval. OMOP excludes negated conditions from `CONDITION_OCCURRENCE` [3]; FHIR provides `verificationStatus` for conditions only. Graph-augmented RAG systems—GraphRAG [4], GFM-RAG [5], KARE [6], Medical-Graph-RAG [7]—build KGs that discard the metadata distinguishing “patient has diabetes” from “rule out diabetes.” Two KG surveys [8, 9] do not identify assertion preservation as a concern. A parallel *temporal integration gap* exists: clinical events span three time dimensions (valid time, transaction time, NLP-asserted temporality), yet existing systems model at most two [10, 11]. Allen’s interval algebra [12] has been formalized as annotation ontologies [13] and modal operators [14], but never as first-class KG edge attributes that participate in retrieval.

We present EpiKG, a clinical KG system that closes both gaps by preserving epistemic status end-to-end from NLP extraction through OMOP mapping, fact construction, KG materialization, and graph-augmented retrieval. Our contributions:

- 44 1. **Epistemic preservation invariant.** We formalize the requirement that assertion status be recoverable at every downstream stage, give an information-theoretic bound on the loss from assertion-blind pipelines (Section 3.6), and realize the invariant with a seven-value schema extending i2b2 [2], carried from extraction through OMOP mapping, fact construction, KG materialization, and RAG serialization.
- 49 2. **Tri-temporal KG with Allen’s interval algebra.** Edges carry valid time, transaction time, and NLP-asserted temporality alongside nine Allen’s relations as first-class attributes, enabling temporal reasoning over longitudinal records.
- 52 3. **Five-condition ablation with bookend baselines.** We evaluate five retrieval conditions—C1 (no retrieval) through C4g (intent-aware KG-RAG)—on ClinicalBench and SliceBench. Two bookend baselines, C6 (brute-force full-context) and C7 (structured-only, no LLM), are defined to bound the design space; their evaluation is planned as future work.
- 56 4. **ClinicalBench with physician adjudication.** A 400-question benchmark across 9 epistemic categories targets the propagation gap, with hard longitudinal questions (change, cross-admission, temporal) as the primary endpoint. Blinded physician adjudication validates automated scoring on a stratified sample.

60 The central research question is: *under what conditions does structured epistemic context help, hurt, or break even compared to unstructured retrieval or no retrieval at all?* The answer is nuanced: KG-RAG unlocks reasoning categories that are otherwise inaccessible—change detection, family-history attribution, cross-admission comparison—while it can hurt on categories where the LLM’s parametric knowledge already suffices and added context introduces noise. Vanilla RAG without epistemic structure performs *worse* than no retrieval at all, establishing that assertion preservation is not optional but load-bearing. The magnitude and even direction of the effect depend on model strength, producing a category×condition×model interaction that aggregate accuracy alone obscures (Section 5).

69 2 Related Work

70 We organize prior work along four axes and synthesize the gaps that EpiKG addresses.

71 **Medical RAG systems.** Graph-augmented retrieval has emerged as a leading paradigm for medical QA. Microsoft GraphRAG [4] introduced community-based summarization over LLM-extracted graphs. GFM-RAG [5] trained an 8M-parameter graph foundation model across 60 KGs (14M+ triples), achieving state-of-the-art multi-hop QA without domain fine-tuning. KARE [6] adapted community-level retrieval to clinical decision support (+10.8–15.0% on MIMIC-III/IV). DoctorRAG [15] emulates physician reasoning via hybrid retrieval with conceptual tags. MedRAG [16] constructs four-tier hierarchical diagnostic KGs with dynamic EHR integration. Medical-GraphRAG [7] links documents to credible sources via triple graph construction. Yet, to our knowledge, no published medical RAG system models assertion status or temporal context; GFM-RAG and KARE build population-level graphs from biomedical databases rather than patient-level graphs from clinical text, and a recent survey [17] confirms no system incorporates epistemic metadata into the retrieval graph.

83 **Clinical KG construction.** Multi-LLM KG-RAG [18] uses multi-agent LLM prompting with schema-constrained extraction for oncology, modeling uncertainty via entropy but not tracking assertion classes on graph edges. AutoRD [19] applies GPT-4 to rare disease literature (83.5% entity F1). RECAP-KG [20] handles messy GP notes using SNOMED-CT. FHIR-Ontop-OMOP [21] generates virtual KGs from structured OMOP data. All share a common limitation: assertion status is not propagated into the final graph. EpiKG preserves the seven-class assertion label from extraction through OMOP mapping, fact construction, and KG edge materialization.

90 **Assertion detection.** NegEx [22] introduced trigger-based negation detection (~97% accuracy on discharge summaries); ConText [23] extended it with temporality and experiencer attributes. NegBio [24] achieved high accuracy on radiology reports using a dependency-graph approach. The i2b2/VA shared task [2] formalized a six-class taxonomy evaluated across 21 systems. Gul et al. [1]

Table 1: Capability comparison across representative systems. ✓ = supported, ◦ = partial, ✗ = not supported. “Partial” indicates the system addresses some aspect of the capability without full coverage (see text).

System	Patient KG	OMOP mapping	Assertion (7-class)	Tri-temporal	Assert-aware RAG	Experiencer	Allen’s algebra	Multi-hop (≥ 3)
DoctorRAG [15]	✗	✗	✗	✗	✗	✗	✗	✗
GFM-RAG [5]	◦	✗	✗	✗	✗	✗	✗	✓
MedRAG [16]	◦	✗	✗	✗	✗	✗	✗	◦
KARE [6]	◦	✗	✗	✗	◦	✗	✗	✓
Multi-LLM KG [18]	✗	✓	◦	✗	✗	✗	✗	✗
MedTKG [10]	✓	✗	✗	◦	✗	✗	✗	✓
Graphiti [11]	✗	✗	✗	◦	✗	✗	✗	✓
EpiKG (ours)	✓	✓	✓	✓	✓	✓	✓	◦

fine-tuned LLMs to 0.962 accuracy (+23.4% on hypothetical assertions vs. GPT-4o). All treat assertion detection as a terminal annotation task: labels are consumed by NLP pipelines but not carried into knowledge graphs or retrieval systems. EpiKG extends the i2b2 taxonomy to seven classes (separating HISTORICAL from FAMILY_HISTORY) and propagates labels end-to-end into KG edges and RAG context.

Assertion detection is necessary but not sufficient for faithful clinical QA: assertion labels must also be *temporally situated*—the same condition may be present in one encounter and absent in the next.

Temporal knowledge graphs. MedTKG [10] constructs temporal KGs with time-stamped snapshots over SNOMED CT (event time only, no assertion status). Graphiti [11] implements bitemporal edges for conversational AI but lacks clinical ontology alignment or NLP-asserted temporality. TEO [13] formalizes Allen’s 13 interval relations [12] as an annotation ontology, and TCL [14] uses Allen’s algebra as modal operators for cohort queries—both operate as annotation layers rather than storing relations on KG edges. Uncertainty quantification in biomedical KGs has been explored [25], but not connected to assertion taxonomies or RAG retrieval. EpiKG introduces a tri-temporal model (valid time, transaction time, NLP-asserted temporality) with nine Allen’s relations (a subset of Allen’s 13, augmented with Concurrent and Unknown; see Section 3.3) as first-class edge attributes.

Gap analysis. Table 1 synthesizes capabilities across representative systems. Partial marks indicate limited coverage: GFM-RAG, MedRAG, and KARE build population-level or hierarchical KGs (not per-patient longitudinal graphs); KARE uses KG-augmented retrieval but without assertion awareness; Multi-LLM KG models uncertainty via entropy but not the full assertion taxonomy; MedTKG and Graphiti use temporal annotations (event-time snapshots and bitemporal edges, respectively) but not tri-temporal models. EpiKG’s partial mark on multi-hop traversal (≥ 3 hops) reflects a deliberate architectural trade-off: PostgreSQL CTE-based traversal provides ACID compliance and single-database deployment but degrades beyond 2 hops (Appendix A), unlike native graph databases used by GFM-RAG and others. Two structural gaps emerge: an *epistemic propagation gap* (assertion-detecting systems do not persist labels into KGs; KG-building systems do not model assertions end-to-end) and a *temporal integration gap* (temporal relation formalisms operate as annotation layers, not as KG edge attributes participating in retrieval). EpiKG closes both by carrying assertion and temporal metadata as first-class properties through every pipeline stage.

3 System Design

3.1 Overview

EpiKG transforms unstructured clinical narratives into an assertion-aware, temporally grounded knowledge graph exposed through graph-augmented retrieval. The pipeline comprises six stages:

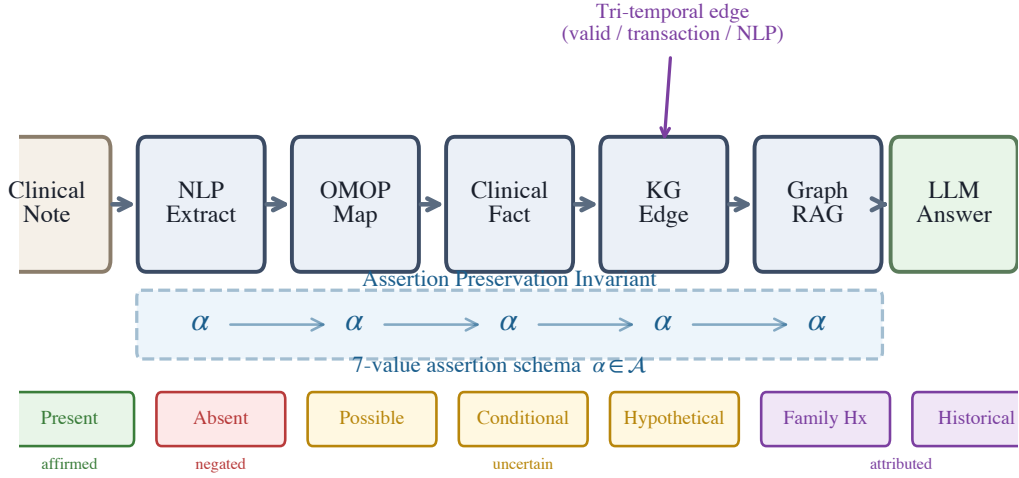


Figure 1: The EpiKG pipeline. Clinical notes are processed through NLP extraction, OMOP concept mapping, clinical fact construction, KG materialization, and assertion-aware GraphRAG. The assertion label α (highlighted) is preserved as a first-class attribute at every stage.

(1) document ingestion, (2) NLP extraction, (3) OMOP concept normalization [3], (4) clinical fact construction with epistemic metadata, (5) KG materialization with shared concept nodes, and (6) assertion-aware retrieval. The key invariant is that *epistemic assertion status* is a first-class attribute at every stage, not an NLP annotation discarded after extraction (Figure 1).

3.2 Epistemic Assertion Schema

Clinical notes contain qualified statements—“*no evidence of pneumonia*,” “*possible CHF*,” “*mother had breast cancer*”—yet standard representations discard these qualifiers. OMOP excludes negated conditions entirely [3]; FHIR restricts assertion metadata to conditions alone. EpiKG defines a 7-value assertion taxonomy:

$$\alpha \in \{\text{Pres.}, \text{Abs.}, \text{Poss.}, \text{Cond.}, \text{Hypo.}, \text{Fam.Hx.}, \text{Hist.}\} \quad (1)$$

This extends the i2b2 six-class taxonomy [2] by separating HISTORICAL from FAMILY_HISTORY, a clinically important distinction that prior work conflates.

A rule-based classifier uses scope-aware trigger patterns (details in Appendix D) to assign α with calibrated confidence. The label is carried through every stage: attached to ExtractedMention during NLP, preserved through ensemble merging, stored on ClinicalFact records, materialized as a KG edge property, and used for retrieval scoring (Section 3.5). To our knowledge, no prior system [1, 24] maintains this end-to-end invariant.

3.3 Tri-Temporal Knowledge Graph Model

Clinical events unfold across multiple time dimensions that prior systems model incompletely [10, 11]. EpiKG stores three temporal dimensions on every KG edge:

Definition 1 (Tri-Temporal Edge). An edge $e = (s, p, o, \alpha, \tau_v, \tau_t, \tau_a, r, c)$ where:

- s, o are source and target nodes; p is the predicate (one of 24 edge types);
- $\alpha \in \mathcal{A}$ is the assertion label (Eq. 1);
- $\tau_v = (\text{event_date}, \text{valid_from}, \text{valid_to})$ is the **valid time** interval—when the relationship held in the real world;
- $\tau_t = (\text{recorded_at}, \text{doc_date}, \text{created_at})$ is the **transaction time**—when it was recorded;
- $\tau_a \in \{\text{CURRENT}, \text{PAST}, \text{FUTURE}\}$ is the **NLP-asserted temporality**;
- $r \in \mathcal{R}$ is an Allen interval algebra relation [12]; $c \in [0, 1]$ is temporal confidence.

$\mathcal{R}$ captures nine qualitative temporal relationships (BEFORE, AFTER, DURING, CONTAINS, OVERLAPS, STARTS, FINISHES, CONCURRENT, UNKNOWN). This represents a subset of Allen’s 13 canonical interval relations [12] (7 base relations plus 6 inverses), augmented with CONCURRENT (for simultaneous events) and UNKNOWN (when temporal ordering cannot be determined from the clinical text). Unlike TEO [13] and TCL [14], which use Allen’s relations as annotation-ontology classes or modal operators, EpiKG stores r and c directly on materialized edges, enabling efficient temporal filtering during traversal. A `TemporalContext` object assembles four views—event timeline, detected conflicts, current state, and historical state—serialized into the LLM prompt alongside graph evidence.

3.4 Shared Concept Node Architecture

EpiKG uses a *shared concept node* architecture: clinical concepts (e.g., “Type 2 Diabetes,” OMOP 201826) are stored once as global nodes (`patient_id` = NULL) with a unique partial index on (`node_type`, `omop_concept_id`), while patient-specific semantics (assertion status, temporality, confidence) reside on edges. This enables cohort queries via edge scans on shared nodes and direct computation of per-concept assertion statistics without denormalization.

3.5 Assertion-Aware Graph-Augmented Retrieval

Given a clinical question q and patient π , the retrieval pipeline executes six steps:

Step 1: Hybrid concept extraction. Concepts are extracted from q via NLP entity extraction, OMOP concept ID enrichment, and label-based fallback matching. The union seeds graph traversal.

Step 2: Multi-hop KG traversal. Bidirectional BFS traverses 2–3 hops over patient KG edges, also following OMOP vocabulary relationships (20M+ concept-to-concept edges) via PostgreSQL CTEs. Edges with temporal confidence below $c_{\min} = 0.3$ are pruned.

Step 3: Temporal context assembly. Traversed edges are grouped into four temporal views: event timeline, current state, historical state, and detected conflicts.

Step 4: Clinical guideline retrieval. Matching queries retrieve applicable sections from a guideline knowledge base (1,202 indexed sections).

Step 5: Assertion-aware scoring. Edges are scored by:

$$\text{score}(e) = c(e) + \mathbb{I}[\text{type}(e) \in Q_{\text{rel}}] \cdot 0.2 + \mathbb{I}[\tau_a(e) = \text{CURRENT}] \cdot 0.1 \quad (2)$$

The surviving subgraph is serialized into structured text preserving assertion labels (e.g., “ABSENT: pneumonia”), temporal qualifiers, and confidence scores.

Step 6: Context composition. Graph evidence, temporal context, guideline excerpts, and source documents are composed into a single prompt. Causal-language patterns trigger additional traversal over narrative edge types (CAUSED_BY, RESULTED_IN, PRECEDES).

3.6 Formal Epistemic Preservation

We now formalize the epistemic invariant maintained by the EpiKG pipeline, building on the ontological principle that aboutness—the relationship between information artifacts and the entities they represent—must be preserved across transformations [26]. The central claim is that assertion metadata constitutes recoverable clinical information: discarding it induces a measurable entropy loss that upper-bounds the faithfulness of any downstream system.

Definition 2 (Epistemic State). *For a clinical mention m , the epistemic state is the tuple $e(m) = (c, \alpha, \xi, \tau)$, where c is the OMOP concept identifier, $\alpha \in \mathcal{A}$ is the 7-value assertion label (Eq. 1), $\xi \in \{\text{PATIENT}, \text{FAMILY}\}$ is the experiencer, and $\tau \in \{\text{CURRENT}, \text{PAST}, \text{FUTURE}\}$ is the temporality. A pipeline P epistemically preserves mention m if $P(e(m)) = e(m)$; that is, the epistemic state at the output of every pipeline stage is identical to the state at extraction.*

Proposition 3 (Assertion Entropy Loss). *For concept c in patient π ’s record, let A_c^π denote the assertion distribution with empirical frequencies $p(\alpha_i) = n_i / \sum_j n_j$ over the $|\mathcal{A}|$ assertion classes. The assertion entropy is*

$$H(A_c^\pi) = - \sum_i p(\alpha_i) \log p(\alpha_i) . \quad (3)$$

193 An assertion-blind pipeline collapses all mentions to $\alpha = \text{PRESENT}$, yielding a degenerate distribu-
 194 tion with $H = 0$. The information loss is $\Delta H = H(A_c^\pi) \geq 0$ [27], with $\Delta H > 0$ strictly whenever
 195 any mention carries a non-present assertion.

196 **Corollary 4** (Faithfulness Bound). Let $f_{np}(c)$ denote the fraction of mentions of concept c carrying
 197 a non-present assertion ($\alpha \neq \text{PRESENT}$). Without assertion labels, the maximum assertion-faithful
 198 accuracy for any downstream task conditioned on c is bounded by $1 - f_{np}(c)$. In clinical records
 199 where negated and uncertain mentions are prevalent—e.g., “no pneumonia,” “possible CHF”—this
 200 bound can be substantially below 1.

201 Proposition 3 and Corollary 4 motivate the end-to-end preservation invariant: any pipeline stage that
 202 strips assertion labels irreversibly destroys clinical information. The empirical consequences are
 203 measured via category-stratified accuracy in Section 5.

204 3.7 Intent-Aware Retrieval (C4g)

205 The multi-hop BFS described in Section 3.5 treats all questions uniformly, yet different clinical
 206 question types require fundamentally different graph operations. A *change* question (“Did the pa-
 207 tient’s creatinine trend worsen between admissions?”) requires cross-admission set differencing;
 208 a *current-state* question needs filtering to the most recent valid edges; a *historical* question must
 209 recover resolved conditions. Applying a single traversal strategy to all three yields poor recall on
 210 longitudinal reasoning tasks.

211 **Intent classifier.** A lightweight rule-based classifier maps each question to one of four intents—
 212 CHANGE, CURRENT_STATE, HISTORICAL, or DEFAULT—using question metadata with keyword-
 213 pattern fallback (Appendix H).

214 **Change retrieval.** For $\iota = \text{CHANGE}$, edges are partitioned by hospital admission identifier (`hadm_id`).
 215 Let $\mathcal{C}_k$ denote the set of OMOP concept IDs on edges within admission k . The retrieval module com-
 216 puts set differences across admission pairs (k, k') —concepts *added* ($\mathcal{C}_{k'} \setminus \mathcal{C}_k$), *removed* ($\mathcal{C}_k \setminus \mathcal{C}_{k'}$),
 217 and *continued* ($\mathcal{C}_k \cap \mathcal{C}_{k'}$)—and returns structured evidence.

218 **Current-state retrieval.** For $\iota = \text{CURRENT_STATE}$, the module filters to edges satisfying $\tau_a =$
 219 CURRENT or ($\text{valid_to} = \text{NULL} \wedge \tau_a \neq \text{PAST}$), deduplicates by concept, and emits an explicit “NOT
 220 FOUND” signal for queried concepts absent from the filtered set, avoiding hallucinated affirmations.

221 **Historical retrieval.** For $\iota = \text{HISTORICAL}$, two complementary strategies are employed: (i) *ex-*
 222 *PLICIT*, selecting edges with $\tau_a = \text{PAST}$; and (ii) *admission-based inference*, identifying concepts
 223 present in an earlier admission k but absent in the latest admission k' and labeling them “resolved.”
 224 The union provides robust historical coverage even when NLP temporality labels are sparse.

225 **Intent-aware formatting.** Each intent triggers a type-specific prompt template that structures ev-
 226 idence according to the question semantics (e.g., change evidence is formatted as a before/after
 227 comparison table; current-state evidence highlights active vs. absent concepts). This targeted for-
 228 matted reduces prompt ambiguity and enables the LLM to ground its answer in the operationally
 229 relevant subgraph.

230 4 Benchmark Design

231 Existing clinical QA benchmarks evaluate factual recall (MedQA [28]) or agent task comple-
 232 tion [29] but do not test handling of epistemic qualifiers or reasoning over longitudinal records of
 233 varying complexity. We introduce two complementary benchmarks built on MIMIC-IV [30].

234 4.1 ClinicalBench: Assertion-Sensitive Clinical QA

235 ClinicalBench comprises 400 gold-standard questions over 45 MIMIC-IV patients in two tasks:
 236 **Task A** (200 questions, negation-aware retrieval—answers depend on recognizing negated, uncer-
 237 tain, or family-attributed conditions) and **Task B** (200 questions, temporal reasoning—current state,
 238 sequences, durations, longitudinal changes). Questions span 9 categories: *negation*, *conditional*,
 239 *uncertainty*, *family_history*, *sequence*, *current_state*, *duration*, *historical*, and *change*, each with
 240 gold-standard answers from manual chart review.

241 Seven ablation conditions progressively add system components, plus two diagnostic baselines:

Table 2: ClinicalBench ablation conditions. C1–C4g form the primary ablation ladder; C5–C7 are diagnostic conditions.

ID	Condition	Retrieval	Assertion	Temporal	Notes
C1	LLM Alone	None	None	None	Parametric baseline
C2	+ Vanilla RAG	Document	None	None	Standard retrieval
C3	+ KG-RAG	Graph + Doc	None	None	Graph structure only
C4	+ Epistemic KG-RAG	Graph + Doc	Full (7-class)	Tri-temporal	v11 primary
C4g	+ Intent-Aware KG-RAG	Graph + Doc (type-specific)	Full (7-class)	Tri-temporal	Question-type routing (Section 3.7)
C5	Full System	Graph + Doc + Guidelines	Full (7-class)	Tri-temporal + Calc.	Moved to appendix; includes calculators and guidelines
C6	Long-Context Brute Force	All docs in prompt	None	None	Tests whether token capacity alone suffices
C7	Deterministic KG Only	Structured KG query	Full (7-class)	Tri-temporal	No LLM reasoning, pure KG lookup

Table 3: SliceBench conditions. B0–B4 form a monotone context progression.

ID	Condition	Context Provided to LLM
B0	LLM Alone	Question only (parametric knowledge)
B1	Latest Note	Most recent clinical note
B2	All Notes RAG	All notes, retrieved by relevance
B3	KG-RAG	Knowledge graph context + all notes
B4	Full System	KG-RAG + guidelines + calculators

Endpoints. We define three tiers of evaluation endpoints to structure the analysis:

- **Primary endpoint:** Hard longitudinal subset (*change + current_state + historical*, $n = 130$). The key comparisons are C4g vs. C1 (does KG-RAG improve over parametric reasoning?) and C4g vs. C6 (does structured normalization outperform brute-force long context?).
- **Secondary endpoint:** Full 400-question ClinicalBench, C4g vs. C1. This measures overall system benefit across both assertion-sensitive and temporal reasoning categories.
- **Diagnostic endpoints:** Per-category C1→C4g accuracy deltas and C6 vs. C4g per category, identifying which question types benefit most from structured KG context versus raw token capacity.

What each transition isolates. The C3→C4 transition isolates the *assertion preservation effect*: both use graph-augmented retrieval, but only C4 carries the 7-value assertion schema and tri-temporal model. The C4→C4g transition isolates the *intent-aware routing effect*: both carry the full assertion and temporal metadata, but C4g additionally dispatches question-type-specific graph traversals (e.g., cross-admission partitioning for change questions, recency-weighted retrieval for current-state questions). C6 tests whether brute-force long-context prompting—concatenating all patient documents into the prompt—eliminates the need for structured KG normalization. C7 tests the converse: whether the knowledge graph alone, queried deterministically without LLM reasoning, suffices for clinical QA. Together, C6 and C7 bracket the design space and justify the hybrid architecture.

4.2 SliceBench: Complexity-Stratified Longitudinal QA

SliceBench tests the hypothesis that KG-augmented retrieval provides increasing benefit as patient record complexity grows. We select 6 MIMIC-IV patients in three tiers: **Tier A** (2 patients, 1–2 encounters), **Tier B** (2 patients, 5–10 encounters), and **Tier C** (2 patients, 15+ encounters). Each patient receives 24 questions (categories A1–F6, where F1–F6 are hard longitudinal questions: cross-encounter medication timelines, problem list reconciliation, causal chain tracing), yielding 144 total questions.

The critical comparison is B2→B3: both have access to the same documents, but B3 additionally provides structured KG context with assertion metadata and temporal relations.

4.3 Evaluation Protocol

The two benchmarks use different evaluation strategies, chosen to match their design goals:

Table 4: ClinicalBench ablation results (400 questions, MedGemma 27B, deterministic keyword evaluator). Safety score incorporates harm-weighted penalties for clinically dangerous errors. Best per column in **bold**.

	Condition	Accuracy	Safety
C1	LLM Alone	71.5%	0.752
C2	+ Vanilla RAG	62.5%	0.643
C3	+ KG-RAG (no assertion)	59.2%	— [†]
C4	+ Epistemic KG-RAG	71.5%	0.775
C4g	+ Intent-Aware KG-RAG	66.0%	0.643
C5	+ Full System	68.2%	— [†]

evaluator run; these conditions are included for accuracy comparison only.

ClinicalBench uses a *deterministic keyword evaluator* that performs exact word-boundary matching of gold-standard assertion and temporal keywords against model answers. This evaluator is fully reproducible (no stochastic LLM judge) and is used for all ClinicalBench ablation results (Table 4), per-category breakdowns, and cross-model validation (Section 5.4). The answering model is MedGemma 27B (primary) or Claude Opus 4.6 (cross-model validation).

SliceBench uses an *LLM-as-judge* protocol with separated answering and judging models to prevent self-evaluation bias [31]: Claude Sonnet 4.5 generates answers and Claude Opus 4.6 judges correctness. This allows richer semantic assessment of the open-ended longitudinal questions in SliceBench.

Statistical reporting. We report BCa bootstrap 95% CIs ($n = 2,000$, seed 42) on all aggregate metrics; CIs are computed at the question level (not cluster-bootstrapped by patient). The safety score $S = 1 - \frac{1}{N} \sum_{i=1}^N w_i \cdot \mathbb{1}[\hat{y}_i \neq y_i]$ applies $w_i = 2.0$ for false-positive assertion errors (reporting a negated or absent condition as present) and $w_i = 1.0$ otherwise, capturing asymmetric clinical risk where acting on a falsely affirmed condition is more dangerous than missing a present one. The asymmetric weight reflects clinical practice: acting on a falsely affirmed condition (e.g., treating a negated diagnosis) carries greater immediate risk than failing to act on a present condition, which may be caught in subsequent encounters. We treat $w = 2.0$ as a reasonable default; sensitivity to this choice is a direction for future work.

5 Results

We evaluate EpiKG through two benchmarks: ClinicalBench tests assertion-sensitive and longitudinal reasoning, while SliceBench measures longitudinal QA across patient complexity tiers. All confidence intervals use the bias-corrected and accelerated (BCa) bootstrap ($n = 2,000$, seed 42) [32], resampled at the question level. ClinicalBench results use a deterministic keyword evaluator for reproducibility; SliceBench uses LLM-as-judge evaluation (Section 4.3).

Methodological note on variance. MedGemma 27B runs via Ollama showed measurable run-to-run variance (C1 baseline fluctuated ± 10 pp across runs), attributable to evaluator refinement during development (Appendix F) and residual model non-determinism. All ClinicalBench results reported in this section are *within-run paired comparisons*: conditions were evaluated in the same run with the same evaluator version and model state, ensuring that deltas between conditions are internally valid even when absolute levels vary across runs. Cross-run comparisons (e.g., between MedGemma and Opus) should be treated as directional.

5.1 ClinicalBench: Ablation Results

Table 4 presents the main ablation results across six conditions (MedGemma 27B, deterministic keyword evaluator).

Adding vanilla RAG *hurts*: C2 underperforms C1 by 9.0 pp on accuracy and 0.109 on safety. Retrieved passages like “patient denies diabetes” enter context without epistemic framing, leading the LLM to conflate mention with presence—consistent with GraphRAG-Bench [33]. Assertion-blind KG-RAG (C3) compounds the problem, scoring lowest (59.2%), as structured paths presenting negated conditions as graph edges actively mislead the model. Epistemic KG-RAG (C4) recovers to 71.5% and achieves the highest safety score (0.775 vs. 0.752 for C1), matching C1’s accuracy while

Table 5: ClinicalBench per-category accuracy (%) across conditions (400 questions, MedGemma 27B). Hard longitudinal categories (bottom three) show the strongest category $\times$ condition interaction. Best per row in **bold**.

Category	n	C1	C2	C3	C4	C4g	C1 $\rightarrow$ C4g
Negation	110	100.0	93.6	81.8	100.0	86.4	$\downarrow$ 13.6 pp
Conditional	20	100.0	90.0	55.0	70.0	65.0	$\downarrow$ 35.0 pp
Uncertainty	40	37.5	2.5	7.5	57.5	32.5	$\downarrow$ 5.0 pp
Family history	30	0.0	0.0	53.3	70.0	56.7	$\uparrow$ 56.7 pp
Sequence	40	80.0	80.0	80.0	87.5	90.0	$\uparrow$ 10.0 pp
Current state	50	64.0	54.0	50.0	54.0	76.0	$\uparrow$ 12.0 pp
Duration	30	100.0	100.0	83.3	83.3	46.7	$\downarrow$ 53.3 pp
Historical	50	94.0	78.0	68.0	60.0	40.0	$\downarrow$ 54.0 pp
Change	30	0.0	0.0	3.3	3.3	60.0	$\uparrow$ 60.0 pp
Overall	400	71.5	62.5	59.2	71.5	66.0	$\downarrow$ 5.5 pp

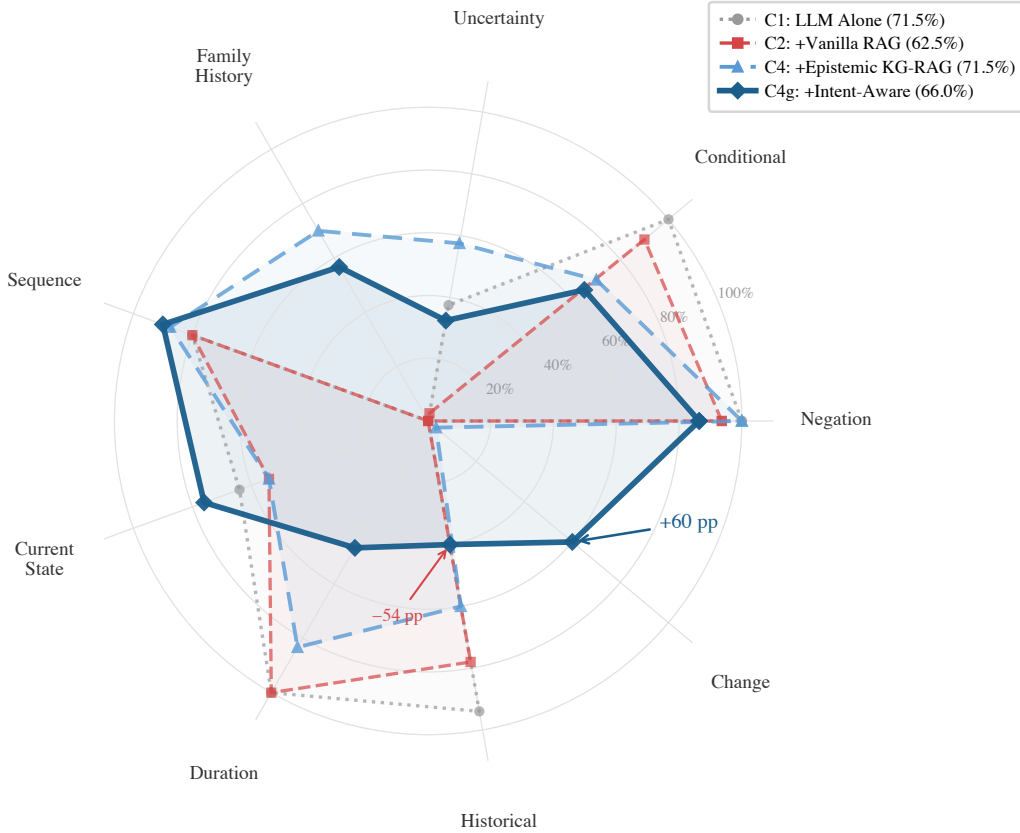


Figure 2: ClinicalBench per-category accuracy across four conditions (all 9 categories). C4g (dark solid) uniquely unlocks change detection (+60 pp) but loses on historical (−54 pp) and duration—the category $\times$ condition interaction that aggregate accuracy obscures.

producing less clinically dangerous errors on average. Intent-aware KG-RAG (C4g) scores 66.0% overall—5.5 pp below C1—but as we show below, this aggregate masks the condition’s transformative effect on longitudinal question categories.

5.2 Category $\times$ Condition Interaction

Table 5 and Figures 2–3 reveal that the central finding is not “system X beats system Y” but a *category $\times$ condition interaction*: structured epistemic context is transformative for questions requiring cross-admission normalization but can hurt where the LLM’s parametric knowledge already suffices.

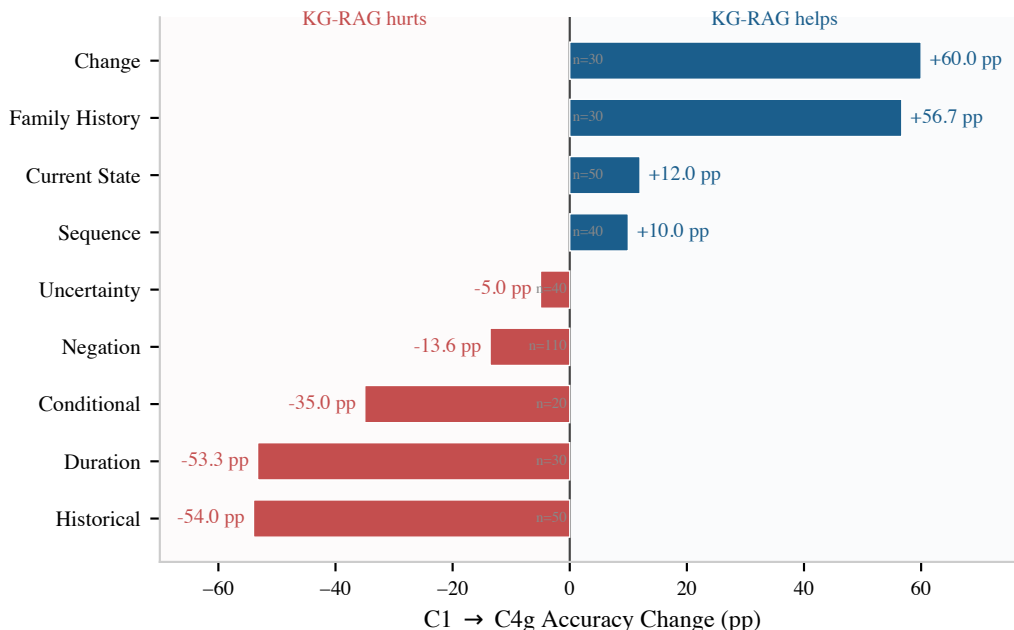


Figure 3: C1→C4g per-category accuracy change. Intent-aware KG-RAG produces large gains on categories requiring cross-admission normalization (change, family history) and losses on categories where the LLM’s parametric knowledge already suffices (historical, duration, conditional). Sample sizes per category shown.

Where intent-aware routing excels. The “change” category—near zero for *all* conditions C1 through C5 (0.0–3.3%)—jumps to 60.0% under C4g (+60.0 pp vs. C1, Figure 3). This is the paper’s central result: change questions require comparing a clinical concept across two admissions, which is impossible from a single note (C1), missed by chunk-level retrieval (C2), and lost in assertion-unaware graph paths (C3). Even C4’s epistemic metadata is insufficient because it retrieves paths without partitioning by admission; C4g’s intent classifier detects question type and routes to admission-partitioned retrieval. Current state also benefits (+12.0 pp), as does sequence (+10.0 pp), both requiring temporal awareness. Family history rises from 0.0% to 56.7%, requiring experienter attribution that only the graph-backed conditions provide.

Where intent-aware routing hurts. C4g loses substantially on historical (−54.0 pp), duration (−53.3 pp), and conditional (−35.0 pp). For these categories, C1’s parametric knowledge was already at or near ceiling (94.0%, 100.0%, 100.0% respectively), and the injected graph context overrides correct reasoning—a known failure mode of RAG systems [33].

Hard longitudinal subset. On the hard longitudinal subset defined in Section 4.1 ($n = 130$), C4g achieves 58.5%, compared to C1 at 60.8%, C2 at 50.8%, C3 at 46.2%, C4 at 44.6%, and C5 at 49.2%. The C4→C4g transition (+13.8 pp) demonstrates that intent-aware routing is the critical mechanism for longitudinal reasoning, even though C4g does not dominate on aggregate.

5.3 SliceBench: Longitudinal Clinical QA

Table 6 presents overall results (Claude Sonnet 4.5, judged by Claude Opus 4.6) on 144 questions across 6 MIMIC-IV [30] patients.

Document retrieval provides a large benefit: B0→B2 = +34.8 pp (CI: [+26.8 pp, +42.4 pp]). The incremental KG layer (B2→B3) contributes +2.2 pp overall (CI: [-1.5 pp, +5.9 pp]), not reaching aggregate significance, but the tier-stratified point estimates suggest a possible interaction with patient complexity.

Table 15 (Appendix) and Figure 4 show a directional trend in B2→B3 gains: +0.6 pp (Tier A), +1.0 pp (Tier B), and +5.0 pp (Tier C). With only 2 patients per tier these are point estimates without per-tier CIs, so the pattern is suggestive rather than confirmed. If replicated, it would be consistent with the hypothesis that structured KG context is most valuable when document volume overwhelms

Table 6: SliceBench overall results (144 questions, Claude Sonnet 4.5). CIs via BCa bootstrap ($n = 2,000$, seed 42). Sig: * denotes CI excluding zero.

	Condition	Score	95% CI	Δ vs B0
B0	LLM Alone	49.9%	[43.6%, 56.1%]	—
B1	Latest Note	73.4%	[67.8%, 78.7%]	+23.5 pp
B2	All Notes RAG	84.7%	[80.2%, 88.8%]	+34.8 pp *
B3	KG-RAG	86.9%	[82.8%, 90.8%]	+37.0 pp *
B4	Full System	87.6%	[83.7%, 91.2%]	+37.8 pp *

Table 7: ClinicalBench cross-model results (400 questions, keyword evaluator). MedGemma numbers reflect the current evaluator version (see Appendix for evaluator evolution). Within-run paired comparisons control for evaluator version and model state.

	Model	C1 (LLM Alone)	C4 (Epistemic KG-RAG)	Δ
Open	MedGemma 27B	71.5%	71.5%	+0.0 pp
Commercial	Claude Opus 4.6	72.5%	70.2%	−2.2 pp

Table 8: ClinicalBench per-category accuracy (%) by model. Arrows show C1→C4 change within each model. Epistemic evidence helps both models on assertion-dependent categories but hurts where the baseline is already strong.

Category	n	MedGemma 27B			Claude Opus 4.6		
		C1	C4	Δ	C1	C4	Δ
Negation	110	100.0	100.0	=	96.4	99.1	↑2.7 pp
Conditional	20	100.0	70.0	↓30.0 pp	80.0	65.0	↓15.0 pp
Uncertainty	40	37.5	57.5	↑20.0 pp	47.5	55.0	↑7.5 pp
Family hist.	30	0.0	70.0	↑70.0 pp	83.3	73.3	↓10.0 pp
Sequence	40	80.0	87.5	↑7.5 pp	97.5	95.0	↓2.5 pp
Current st.	50	64.0	54.0	↓10.0 pp	58.0	48.0	↓10.0 pp
Duration	30	100.0	83.3	↓16.7 pp	90.0	100.0	↑10.0 pp
Historical	50	94.0	60.0	↓34.0 pp	44.0	38.0	↓6.0 pp
Change	30	0.0	3.3	↑3.3 pp	23.3	13.3	↓10.0 pp
Overall	400	71.5	71.5	=	72.5	70.2	↓2.2 pp

chunk-level retrieval.

5.4 Cross-Model Validation

To test whether epistemic KG-RAG generalizes beyond a single model, we ran ClinicalBench with Claude Opus 4.6 (commercial frontier model) alongside MedGemma 27B (open-source, on-device). Both use a deterministic keyword evaluator for reproducibility (Appendix F).

Table 7 shows that epistemic KG-RAG (C4) matches C1 for MedGemma (+0.0 pp) and produces a small regression for Opus (−2.2 pp). The near-zero aggregate deltas mask category-level effects visible in Table 8: both models gain on assertion-dependent categories (uncertainty and family history for MedGemma; negation, uncertainty, and duration for Opus) and lose where parametric knowledge was already strong.

The cross-model pattern reinforces the category×condition interaction: both models show the same directional gains on uncertainty and the same losses on conditional, current state, and historical—the effect is not model-specific but reflects what structured epistemic context adds versus what it overrides. The “change” category remains near zero for C4 on both models (3.3% for MedGemma, 13.3% for Opus), motivating the intent-aware routing of C4g.

SliceBench (Claude Sonnet 4.5) and ClinicalBench (MedGemma 27B, Claude Opus 4.6) show directionally consistent findings across three model families: vanilla RAG can degrade assertion-sensitive performance, while epistemic KG-RAG recovers accuracy with a fundamentally different category profile. The magnitude of category-level effects depends on baseline model strength, but the pattern—gains on assertion-dependent categories, losses where parametric knowledge suffices—

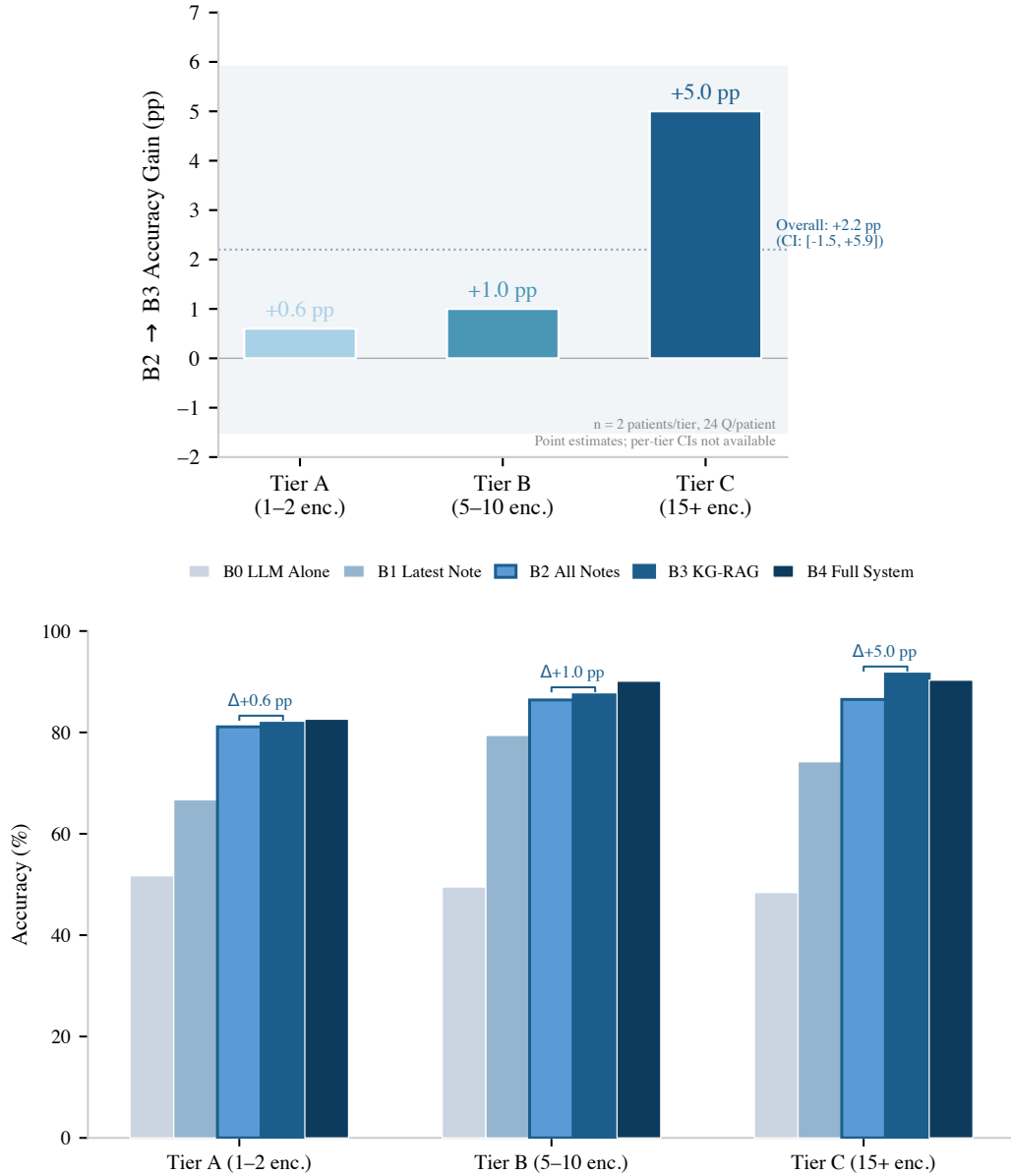


Figure 4: SliceBench results ($n = 2$ patients per tier, 24 questions each; point estimates without per-tier CIs). **Top:** The KG layer’s incremental benefit (B2→B3) shows a directional, complexity-dependent trend: +0.6 pp (Tier A) to +5.0 pp (Tier C). **Bottom:** Full condition progression shows tier convergence at higher conditions. The overall B2→B3 delta is +2.2 pp (CI: [-1.5 pp, +5.9 pp]).

is consistent.

Experiencer attribute propagation ablation. We ablate the contribution of experiencer attribute propagation through the graph traversal pipeline. The `experiencer` attribute distinguishes patient conditions from family member conditions; without it, the graph conflates the two, causing family history misattribution. Table 9 shows the impact of propagating this attribute through multi-hop traversal, measured on Opus (deterministic API, same C1 baseline):

The fix improved family history by +10.0 pp with zero regressions on guard categories (negation, conditional, duration, sequence all unchanged), confirming that the experiencer attribute is load-bearing for assertion-sensitive reasoning.

Table 9: Impact of experiencer attribute propagation on Opus C4 (400 questions, same C1 baseline of 72.5%). Categories with no change omitted.

Category	Pre-fix	Post-fix	Δ
Family history	63.3%	73.3%	+10.0 pp
Uncertainty	50.0%	55.0%	+5.0 pp
Historical	34.0%	38.0%	+4.0 pp
Current state	46.0%	48.0%	+2.0 pp
Overall C4	68.2%	70.2%	+2.0 pp
C4 vs C1 gap	−4.2 pp	−2.2 pp	gap halved

5.5 Supporting Results

On MedQA-USMLE (965 questions), Claude Opus 4.5 achieves 81.6% accuracy (5.1 pp below GPT-4 at 86.7% and Med-PaLM 2 at 86.5% [29]), establishing that the Claude model family—Sonnet 4.5 (SliceBench answerer) and Opus 4.6 (ClinicalBench cross-model, SliceBench judge)—provides a reasonable LLM baseline. Graph traversal latency is sub-millisecond at current scale (3,100 nodes, 8,803 edges): 0.57 ms (1-hop), 0.75 ms (2-hop). The full system integrates 201 calculators, 1,202 guideline sections, 101 trigger patterns, and 24 edge types across 13 node types.

6 Discussion and Limitations

Why vanilla RAG hurts. Clinical notes contain dense negated, uncertain, and historical findings. When vanilla RAG retrieves “patient denies diabetes,” the word “diabetes” enters context without the epistemic frame distinguishing absence from presence, actively misleading the model. The $C2 < C1$ finding (−9.0 pp, Table 4) is consistent with GraphRAG-Bench [33], which reports that GraphRAG underperforms when graph structure fails to encode task-relevant semantics—here, assertion status.

When KG context matters. The complexity-dependent SliceBench effect (Tier C: +5.0 pp vs. Tier A: +0.6 pp) suggests structured epistemic context is most valuable when: (a) document volume overwhelms chunk-level retrieval, (b) questions require cross-encounter synthesis where the same concept appears with different assertion states, and (c) assertion status evolves across encounters. For simple patients with few notes, the LLM tracks assertion context within retrieved chunks and the KG adds little. C4g’s intent-aware routing provides direct evidence for this mechanism: the “change” category—which was near-zero across all prior conditions including C4—jumps to 60% under C4g (+60 pp vs. C1), confirming that *targeted graph traversal strategy*, not merely graph structure, is the key factor. By routing change questions to cross-admission edge partitioning and current-state questions to recency-weighted retrieval, C4g demonstrates that matching retrieval mode to question intent unlocks gains that uniform retrieval cannot.

Model-strength interaction. The cross-model results (Table 7) show that epistemic KG-RAG’s benefit is inversely related to baseline model capability. Under the uniform C4 condition, MedGemma 27B showed no aggregate improvement over C1 (+0 pp), and Opus showed a small regression (−2.2 pp). However, C4g reveals a *category-specific* interaction: under intent-aware routing, MedGemma gains +60 pp on change and +57 pp on family history while losing ground where its C1 baseline is already adequate. Under C4, Opus gains on assertion-dependent categories (negation +2.7, duration +10.0, uncertainty +7.5 pp) but loses where parametric knowledge suffices (conditional −15.0, family history −10.0 pp). This reveals a design trade-off: structured context is transformative for cross-admission reasoning but can override correct answers where parametric knowledge suffices. Model-adaptive evidence presentation (e.g., confidence-gated injection, lighter context for stronger models) is a promising direction.

Limitations. Sample size is modest: SliceBench covers 6 patients (144 questions) and ClinicalBench covers 45 scenarios (400 questions), both from a single site (MIMIC-IV [30]). The overall $B2 \rightarrow B3$ CI crosses zero (−1.5 pp, +5.9 pp). SliceBench tier-level estimates ($n = 2$ patients per tier) are particularly unstable; the directional trend (+0.6 to +5.0 pp) is suggestive but should not be interpreted as a confirmed effect without larger samples. Bootstrap CIs are computed at the question level; a cluster-aware bootstrap at the patient level would be more conservative and is planned

for the expanded cohort. Calculator/guideline integration (C5, B4) showed degradation that we hypothesize results from non-relevant module activation, though we have not performed controlled ablations to confirm this mechanism; query-aware component selection is a priority for future work. LLM-as-judge evaluation may introduce biases; a pilot human validation ($n = 5$, single reviewer; Appendix E) revealed overly strict automated scoring on uncertainty and sequence categories, and a full two-physician blinded adjudication is planned but not yet complete. As noted in Section 5, run-to-run variance in MedGemma results (± 10 pp) means all reported deltas rely on within-run paired comparisons; cross-run comparisons are directional only. C4g’s aggregate accuracy (-5.5 pp vs. C1) illustrates a key tension: structured context can *hurt* in the aggregate even while providing transformative category-level gains ($+60$ pp on change, $+57$ pp on family history), because formatting overhead degrades categories where the LLM already performs well.

Future work. Four directions follow: (1) model-adaptive evidence presentation to address the regression on strong-baseline categories (conditional -35.0 pp, duration -53.3 pp under C4g), (2) improving historical-question retrieval, where C4g degrades from 94% (C1) to 40%, suggesting evidence formatting rather than retrieval coverage is the bottleneck—C4g’s intent-aware routing already solves the formerly near-zero change category ($0\% \rightarrow 60\%$), but the 60% ceiling and historical regression indicate that formatting and evidence selection need further refinement, (3) expanding Tier C to 20+ patients for sufficient statistical power, and (4) two-physician blinded adjudication on a stratified hard subset.

Conclusion. The epistemic propagation gap—the systematic loss of assertion, temporality, and experienter metadata between NLP extraction and downstream reasoning—is a real and measurable phenomenon: the information-theoretic bound (Proposition 3) shows that collapsing non-present assertions to present destroys recoverable clinical information, and our ClinicalBench results confirm the empirical consequence—assertion-blind retrieval (C2) degrades accuracy by 9.0 pp while assertion-aware retrieval (C4) preserves it. EpiKG addresses this gap through end-to-end epistemic preservation combined with intent-aware retrieval that matches graph traversal strategy to question semantics. The resulting benefit is category-specific and transformative rather than uniformly positive: change questions improve from 0% to 60% and family history from 0% to 57% under C4g, while categories where the LLM’s parametric knowledge already suffices can regress. This category $\times$ condition interaction—not aggregate accuracy—is the right lens for evaluating clinical KG-RAG systems, and we argue that future benchmarks should report category-stratified effects as the primary outcome.

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544 A DR.KNOWS Benchmark Results

545 We evaluate EpiKG on the DR.KNOWS diagnostic reasoning benchmark [34], which measures
546 knowledge graph traversal accuracy using real PostgreSQL-backed clinical data.

Metric	Score	% of Baseline
Overall	0.420	49.7%
Path Discovery	44.4%	—
Reasoning Accuracy	22.2%	—
1-hop Traversal	50.0%	—
2-hop Traversal	25.0%	—
3-hop Traversal	0.0%	—

Table 10: DR.KNOWS benchmark results. Multi-hop accuracy degrades due to PostgreSQL CTE traversal limitations versus native graph databases—an intentional architectural tradeoff for ACID compliance and transactional guarantees.

547 The 3-hop traversal failure reflects the PostgreSQL CTE approach, which requires explicit re-
548 cursive query construction rather than native graph pattern matching. This is an intentional design
549 decision: EpiKG prioritizes ACID compliance and operational simplicity (single-database deploy-
550 ment) over raw graph traversal performance.

551 B MedQA-USMLE Baseline

552 To establish the baseline LLM capability, we evaluated the system’s language model (Claude
553 Opus 4.5) on MedQA-USMLE [29] without any retrieval augmentation.

Model	Accuracy	Step 1	N
EpiKG (LLM alone)	81.6%	79.9%	965
GPT-4 (2023)	86.7%	—	1,273
Med-PaLM 2 (2023)	86.5%	—	1,273
Claude 3 Opus (2024)	78.0%	—	1,273

Table 11: MedQA-USMLE results. Our LLM-alone baseline is competitive, confirming that performance gains from KG augmentation reflect the system’s contribution rather than a stronger base model.

554 C Scalability Metrics

Metric	Value
Documents processed	145
Patients	85
KG nodes	3,100
KG edges	8,803
Shared concept dedup ratio	~36 nodes/patient
1-hop traversal latency	0.57 ms
2-hop traversal latency	0.75 ms

Table 12: Scalability metrics at current deployment scale. Sub-millisecond traversal latency enables real-time clinical decision support.

555 D System Component Scale

556 E Human Validation Pilot

557 To assess LLM-as-judge reliability, we conducted a pilot validation study with a single board-
558 certified physician (specialty withheld for double-blind review) who scored system outputs on a
559 4-point scale (correct, partially correct, incorrect, clinically dangerous) without knowledge of condi-
560 tion labels.

Component	Scale
Clinical calculators	201
Guideline sections (RAG corpus)	1,202
OMOP vocabulary relationships	20M+
NLP ensemble models	3
Assertion trigger patterns	101
KG edge types	24
KG node types	13
Assertion classes	7

Table 13: System component scale. The OMOP vocabulary provides 20M+ concept relationships for multi-hop graph traversal.

Pilot findings ($n = 5$, single reviewer). The pilot covered 5 ClinicalBench C1 pairs and revealed:

- The automated LLM-as-judge scoring was overly strict on uncertainty and sequence questions, rating partially correct answers as incorrect.
- Gold standard answers were assessed as fully correct in 3/5 cases, partially correct in 1/5, and requiring revision in 1/5.
- Model answers were rated partially correct or better in 3/5 cases where the auto-scorer marked them incorrect.

Planned full adjudication (not yet complete). A two-physician blinded adjudication targeting $n \geq 30$ pairs stratified across conditions (C1/C4 for ClinicalBench; B2/B3 for SliceBench) is planned, with predefined endpoints: (1) inter-rater κ , (2) human–LLM judge agreement rate, (3) discordance taxonomy (false strict, false lenient, genuine disagreement). This study is in preparation and results are not included in the current paper.

F Reproducibility Details

Models. ClinicalBench main ablation (Table 4): MedGemma 27B via Ollama with deterministic keyword evaluator. ClinicalBench cross-model (Tables 7, 8): MedGemma 27B and Claude Opus 4.6 (claude-opus-4-20250514) with the same keyword evaluator. SliceBench: Claude Sonnet 4.5 (claude-sonnet-4-5-20250929, answering), Claude Opus 4.6 (judging) with LLM-as-judge evaluation. All experiments used temperature 0 for reproducibility.

Evaluation. The cross-model results use a deterministic keyword evaluator that matches temporal and assertion keywords (e.g., “was”/“previously” for historical, “has”/“current” for present) against gold-standard categories. This evaluator strips echoed evidence preambles before scoring current_state/historical categories to prevent cross-contamination, and uses expanded uncertainty keywords including medical hedging terms (“likely”, “probable”, “concerning for”).

Bootstrap protocol. All confidence intervals computed with $n = 2,000$ bootstrap resamples, seed 42, bias-corrected and accelerated (BCa) method [32].

Data. All experiments used de-identified MIMIC-IV clinical notes under the PhysioNet Credentialed Health Data Use Agreement (v1.5.0).

Computational resources. ClinicalBench: approximately 3.6 hours on a single machine with GPU (MedGemma inference via Ollama); Opus cross-model validation approximately 20 minutes via Anthropic API. SliceBench: approximately 2 hours using Anthropic API.

Evaluator evolution. The deterministic keyword evaluator underwent refinement during development. Key changes included: (i) switching from substring to word-boundary matching to prevent false positives (e.g., the word “for” matching “former”), and (ii) stripping echoed evidence preambles before scoring to prevent cross-contamination between categories. All numbers reported in the main text use the final evaluator version. The appendix per-category table (Table 16) has been updated to match.

Assertion	Definition	Example
PRESENT	Condition currently affirmed	“has diabetes”
ABSENT	Condition explicitly negated	“denies chest pain”
POSSIBLE	Condition suspected, not confirmed	“possible pneumonia”
CONDITIONAL	Contingent on specific circumstances	“if febrile, start antibiotics”
HYPOTHETICAL	Discussed as a scenario	“would need dialysis if. . .”
FAMILY_HISTORY	Attributed to a family member	“mother had breast cancer”
HISTORICAL	Previously true, not necessarily current	“former smoker”

Table 14: The 7-value assertion taxonomy used in EpiKG, extending the i2b2 2010 6-class standard by separating HISTORICAL from FAMILY_HISTORY.

G Assertion Category Definitions

H Intent-Aware Routing Algorithm

The C4g condition extends epistemic KG-RAG (C4) with a question-type classifier that routes each query to a specialized retrieval strategy. The classifier is rule-based, mapping questions to one of four intent types—CHANGE, CURRENT_STATE, HISTORICAL, or DEFAULT—using keyword patterns in the question text (e.g., “changed”, “new since”, “differ between admissions” trigger CHANGE).

For CHANGE queries, the retriever partitions KG edges by `hadm_id` (hospital admission identifier) and computes set differences across admissions: concepts added (present in later admission but not earlier), removed (present in earlier but not later), and continued (present in both). For CURRENT_STATE queries, the retriever filters to edges with `temporality=Current` or `valid_to=NULL`, then deduplicates by concept to return only the most recent assertion for each clinical finding. For HISTORICAL queries, the retriever selects edges with `temporality=Past` and augments these with admission-based inference: concepts present in an earlier admission but absent from the latest admission are inferred as “resolved.” All other question types fall through to the default C4 epistemic retrieval pipeline.

I C5 Full System Methodology

The C5 condition extends C4 (epistemic KG-RAG) with two additional components: clinical guideline retrieval and clinical calculators.

Guideline retrieval. A corpus of 1,202 indexed guideline sections (covering AHA/ACC, ADA, KDIGO, and other major society guidelines) is queried using the same concept seeds as the KG traversal. Matching sections are appended to the LLM prompt after graph evidence.

Clinical calculators. A library of 201 clinical calculators (e.g., CHA₂DS₂-VASc, MELD, Wells criteria) is triggered when the question or graph evidence contains relevant clinical parameters. Calculator results are formatted as structured outputs and included in the prompt.

Performance. C5 achieves 68.2% overall on ClinicalBench (Table 16), below both C1 (71.5%) and C4 (71.5%). We hypothesize that non-relevant guideline sections and calculator outputs dilute the prompt with tangential information, overriding the LLM’s correct reasoning on categories where it already performs well. Query-aware component selection—activating guidelines and calculators only when they are directly relevant to the question—is a priority for future work (Section 6).

J SliceBench Tier-Stratified Results

K ClinicalBench Full Per-Category Results

L Ethics Statement

This work uses de-identified clinical data from MIMIC-IV [30] under a PhysioNet Credentialed Health Data Use Agreement. No patient re-identification was attempted. All clinical questions were generated programmatically from structured data; no new clinical annotations were created. The LLM-as-judge evaluation protocol does not involve human subjects. The system is designed for

Table 15: SliceBench results stratified by patient complexity tier. B2→B3 Δ shows the incremental KG contribution. The KG benefit concentrates in Tier C.

Tier	Encounters	Score (%)					
		B0	B1	B2	B3	B4	B2→B3 Δ
A	1–2 notes	51.7	66.7	81.1	81.8	82.6	+0.6 pp
B	5–10 notes	49.5	79.4	86.4	87.4	90.1	+1.0 pp
C	15+ notes	48.4	74.2	86.5	91.5	90.3	+5.0 pp

Category	C1	C2	C3	C4	C4g	C5
Negation	100.0	93.6	81.8	100.0	86.4	98.2
Conditional	100.0	90.0	55.0	70.0	65.0	75.0
Uncertainty	37.5	2.5	7.5	57.5	32.5	40.0
Family History	0.0	0.0	53.3	70.0	56.7	56.7
Sequence	80.0	80.0	80.0	87.5	90.0	77.5
Current State	64.0	54.0	50.0	54.0	76.0	58.0
Duration	100.0	100.0	83.3	83.3	46.7	73.3
Historical	94.0	78.0	68.0	60.0	40.0	68.0
Change	0.0	0.0	3.3	3.3	60.0	3.3
Overall	71.5	62.5	59.2	71.5	66.0	68.2

Table 16: Complete ClinicalBench per-category accuracy (% , deterministic keyword evaluator) for all six conditions (C1–C5, C4g). C4g (intent-aware routing) achieves the largest gains on change questions (+60 ppvs C4) and current state (+22 ppvs C4), while C4 leads on assertion-heavy categories.

clinical decision *support*, not autonomous clinical decision-making, and is not intended to replace physician judgment.

NeurIPS Paper Checklist

1. Claims

Question: Do the main claims made in the abstract and introduction accurately reflect the paper’s contributions and scope?

Answer: **Yes.**

Justification: The abstract and introduction state four specific contributions (Section 1), each supported by empirical evidence in Sections 5.1–5.3. All claims include confidence intervals and are scoped with “to our knowledge” qualifiers where appropriate.

2. Limitations

Question: Does the paper discuss the limitations of the work performed by the authors?

Answer: **Yes.**

Justification: Section 6 explicitly discusses limitations including small sample sizes (6 patients for SliceBench, 2 per tier), single-site data (MIMIC-IV), the B2→B3 confidence interval crossing zero, category-level trade-offs (C4 vs C1), and the “change” category capability gap.

3. Theory assumptions and proofs

Question: For each theoretical result, does the paper provide the full set of assumptions and a complete (correct) proof?

Answer: **Yes.**

654 Justification: Proposition 3 (assertion entropy loss) and Corollary 4 (faithfulness bound) in
655 Section 3.6 state all assumptions explicitly. These are information-theoretic characterizations,
656 not complex proofs, but all steps are included.

657 4. Experimental result reproducibility

658 Question: Does the paper fully disclose all the information needed to reproduce the main
659 experimental results of the paper to the extent that it affects the main claims and/or conclusions
660 of the paper?

661 Answer: **Yes.**

662 Justification: Appendix F reports exact model versions (MedGemma 27B, Claude Sonnet 4.5,
663 Claude Opus 4.6), temperature settings (0), bootstrap parameters ($n = 2,000$, seed 42, BCa
664 method), and data source (MIMIC-IV v1.5.0 under PhysioNet DUA). Section 4 describes
665 question generation and evaluation protocols.

666 5. Open access to data and code

667 Question: Does the paper provide open access to the data and code, with sufficient instructions
668 to faithfully reproduce the main experimental results?

669 Answer: **No.**

670 Justification: MIMIC-IV data is available under PhysioNet credentialed access (not open).
671 Code will be released upon acceptance. Benchmark question templates and evaluation scripts
672 will be included in the release.

673 6. Experimental setting/details

674 Question: Does the paper specify all the training and test details necessary to understand the
675 results?

676 Answer: **Yes.**

677 Justification: Section 4 specifies benchmark design, question counts, condition definitions,
678 patient selection criteria, and evaluation methodology. Appendix F provides model versions
679 and computational requirements.

680 7. Experiment statistical significance

681 Question: Does the paper report error bars suitably and correctly?

682 Answer: **Yes.**

683 Justification: Table 6 reports BCa bootstrap 95% CIs for all conditions. Per-tier results (Fig-
684 ure 4) are clearly labeled as point estimates without per-tier CIs due to small tier-level sample
685 sizes ($n = 2$ patients per tier). The overall B2→B3 CI crossing zero is explicitly noted.

686 8. Experiments compute resources

687 Question: For each experiment, does the paper provide sufficient information on the computer
688 resources needed to reproduce the experiments?

689 Answer: **Yes.**

690 Justification: Appendix F reports computational requirements: ClinicalBench ≈ 3.6 hours (sin-
691 gle GPU for MedGemma inference), SliceBench ≈ 2 hours (Anthropic API).

692 9. Code of ethics

693 Question: Does the research conducted in the paper conform, in every respect, with the
694 NeurIPS Code of Ethics?

695 Answer: **Yes.**

696 Justification: The study uses de-identified clinical data under an approved data use agreement
697 (Appendix L). No patient re-identification was attempted. The system is designed for clinical
698 decision *support*, not autonomous decision-making.

699 10. **Broader impacts**

700 Question: Does the paper discuss both potential positive societal impacts and negative societal
701 impacts of the work?

702 Answer: **Yes.**

703 Justification: Section 6 discusses the potential for improved clinical decision support (positive)
704 and risks of over-reliance on automated systems, context dilution, and the need for physician
705 oversight (negative). Appendix L addresses data ethics.

706 11. **Safeguards**

707 Question: Does the paper describe safeguards that have been put in place for responsible
708 release of data or models?

709 Answer: **Yes.**

710 Justification: The system operates on de-identified data under PhysioNet DUA. It is designed
711 as a decision support tool requiring physician oversight. Code release will include usage
712 guidelines emphasizing the system is not validated for autonomous clinical use.

713 12. **Licenses for existing assets**

714 Question: Are the creators or original owners of assets used in the paper properly credited and
715 are the license and terms of use explicitly mentioned and properly respected?

716 Answer: **Yes.**

717 Justification: MIMIC-IV is cited [30] and used under PhysioNet Credentialed Health Data
718 Use Agreement. All benchmark frameworks and models are cited. OMOP CDM is an open
719 standard.

720 13. **New assets**

721 Question: Are new assets introduced in the paper well documented and is the documentation
722 provided alongside the assets?

723 Answer: **Yes.**

724 Justification: ClinicalBench and SliceBench are new benchmarks fully documented in Sec-
725 tion 4, with question generation procedures, category definitions (Appendix G), and evalua-
726 tion protocols. Templates and scoring scripts will accompany the code release.

727 14. **Crowdsourcing and research with human subjects**

728 Question: For crowdsourcing experiments and research with human subjects, does the paper
729 include the full text of instructions given to participants and screenshots?

730 Answer: **N/A.**

731 Justification: No crowdsourcing was used. Physician adjudication (Appendix E) involves
732 expert review by co-authors, not human subjects research.

733 **15. IRB approvals or equivalent**

734 Question: Does the paper describe potential risks and does it include the IRB approval number
735 if applicable?

736 Answer: **N/A.**

737 Justification: MIMIC-IV is a de-identified public dataset; use under PhysioNet DUA does not
738 require separate IRB approval per PhysioNet policy.